

STAT/MA 41600
In-Class Problem Set #16: September 25, 2026
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Problem Set 16 Answers

1. The probability that they both roll their respective dice x times each is:

$$(5/6)^{x-1}(1/6)(19/20)^{x-1}(1/20) = (19/24)^{x-1}(1/120).$$

So the probability that each of them roll their respective dice the same number of times is $\sum_{x=1}^{\infty} (19/24)^{x-1}(1/120) = \frac{1/120}{1-19/24} = \frac{1/120}{5/24} = 1/25$.

2. We compute as follows: $P(Y \geq X) = \sum_{x=1}^{\infty} \sum_{y=x}^{\infty} (3/5)^{x-1}(2/5)(4/9)^{y-1}(5/9) = (2/9) \sum_{x=1}^{\infty} (3/5)^{x-1} \sum_{y=x}^{\infty} (4/9)^{y-1} = (2/9) \sum_{x=1}^{\infty} (3/5)^{x-1} \frac{(4/9)^{x-1}}{1-4/9} = (2/5) \sum_{x=1}^{\infty} (4/15)^{x-1} = (2/5)/(1-4/15) = (2/5)/(11/15) = 6/11$.

Alternatively, we can think about X as the number of trials until the first success, with probability $2/5$ of success on each trial. We can think about Y as the number of trials (in a separate set of trials) until the first success, with probability $5/9$ of success on each trial. The probability that one or the other or both trials has a success is $(2/5)(5/9) + (2/5)(4/9) + (3/5)(5/9)$, so we conclude that $P(Y \geq X)$ is $\frac{(2/5)(5/9)+(2/5)(4/9)}{(2/5)(5/9)+(2/5)(4/9)+(3/5)(5/9)} = (10+8)/(10+8+15) = 18/33 = 6/11$.

3. Method 1: The total expected number of rolls is $1/(1/6) = 6$. The last roll is always red. So the expected number of rolls that occur *strictly before the red roll* is 5. Each such roll (before the red roll) has 5 equally likely results, two of which are blue and three of which are yellow, and therefore, the expected number of blue results is $(5)(2/5) = 2$.

Method 2: We can simply ignore the yellow sides, so that each trial has 3 equally likely results, namely, red (which is a success, and the trials stop), and two blues (which are failures). With this view, we have a Geometric number of trials with probability of success $p = 1/3$ on each trial, and the expected number of failures is $(1-p)/p = 2$.

Method 3: On each trial, there is $1/6$ probability of stopping, $2/6$ probability of a blue side, and $3/6$ probability of a result that does not matter, so

$$E(X) = (1/6)(0) + (2/6)(1 + E(X)) + (3/6)(E(X)),$$

and therefore $(1 - 2/6 - 3/6)E(X) = (2/6)(1)$, so $E(X) = 2$.

4. We have $P(X \geq 7 | Y \geq X) = \frac{P(Y \geq X \geq 7)}{P(Y \geq X)}$, and the denominator is $6/11$, as calculated in 2. We also have $P(Y \geq X \geq 7) = \sum_{x=7}^{\infty} \sum_{y=x}^{\infty} (3/5)^{x-1}(2/5)(4/9)^{y-1}(5/9) = (2/9) \sum_{x=7}^{\infty} (3/5)^{x-1} \sum_{y=x}^{\infty} (4/9)^{y-1} = (2/9) \sum_{x=7}^{\infty} (3/5)^{x-1} \frac{(4/9)^{x-1}}{1-4/9} = (2/5) \sum_{x=7}^{\infty} (4/15)^{x-1} = (2/5)(4/15)^6/(1-4/15) = (2/5)(4/15)^6/(11/15) = (6/11)(4/15)^6$. So we conclude that $P(X \geq 7 | Y \geq X) = (4/15)^6$.

- 5a. We have $P(X + Y + Z > 10) = \binom{10}{0}(1-p)^{10} + \binom{10}{1}(1-p)^9p + \binom{10}{2}(1-p)^8p^2$.

- 5b. We have $P(X + Y + Z > k) = \binom{k}{0}(1-p)^k + \binom{k}{1}(1-p)^{k-1}p + \binom{k}{2}(1-p)^{k-2}p^2$.